

ANALYSIS

Based on the experimental results obtained from my local trials, below is a detailed analysis of my Multi-Agent AI system's performance and logic.

1. The Role of `worker_responses` and `operator.add`

In my LangGraph state definition, the `expert_outputs` (or `worker_responses`) list acts as a shared accumulator.

- **List Transformation:** As each expert node (Scientist, Critic, Creative) completes its task in the Map phase, it returns a new list containing its specific dictionary of results. The graph then appends these results to the existing list in the state.
- **Essentiality of `operator.add`:** This operator is essential for the Reduce design pattern because the nodes run in parallel. Without `operator.add`, each node's output would overwrite the previous one in the state; instead, this reducer ensures that all three outputs are collected and preserved for the Judge node to process simultaneously.

2. Model Performance: Speed and Depth

- **Fastest Model:** In Trial A, the Creative model (smollm2:1.7b) was the fastest, finishing in 3.33s. In Trial C, the Scientist model (llama3.2:1b) was significantly faster, finishing in just 2.36s.
- **Critique Depth:** The DeepSeek-R1 reasoning model generally provides a more refined critique when it functions correctly. In Trial C, it identifies specific criteria like "originality" and "derivative nature," whereas Llama 1B provides a more basic factual statement. However, in Trial B, DeepSeek-R1 failed to provide a deeper critique because it hit a limit, resulting in "No output — reasoning tokens consumed".

3. Conflict Resolution by the Judge

The Judge node successfully resolved conflicts by identifying common ground. For example, in Trial C, the Scientist argued AI code is not eligible for protection while the Critic argued it hinges on originality.

- Consensus Outcome: The Judge successfully synthesized these by stating the issue is "contentious" and balanced the two opposing viewpoints (originality vs. derivative nature) into a single, cohesive sentence.

4. Memory Pressure and Lag (8GB Machine Observations)

During the Map phase, when all three models are active, significant memory pressure occurs as the system attempts to load multiple model weights into RAM/VRAM simultaneously.

- Observation: You likely observed latencies spiking during the Map phase. In Trial B, the combined activity led to the Critic node exhausting its reasoning tokens. The high latencies for DeepSeek (often exceeding 18-20 seconds) compared to the low latency of the final Judge node (under 5 seconds) indicate a bottleneck when multiple agents are competing for CPU/RAM resources.

5. Handling Inaccuracies and Hallucinations

In Trial A, the Scientist model (llama3.2:1b) provided a somewhat generalized statement that automation is "replacing some jobs," which could be seen as an oversimplification.

- Filtering: The Judge node addressed this by incorporating the Creative node's insight that AI also "creates new job opportunities" and "enhances productivity". The Judge filtered out the narrower "replacement" narrative from the Scientist and created a more balanced consensus that included both positive and negative changes.

6. Execution Time Comparison

The execution times varied significantly based on model complexity:

- Comparison: DeepSeek-R1 consistently took the longest (e.g., 18.83s and 20.43s) while Llama and SmolLM remained under 18s and sometimes as low as 2.36s.
- Reasoning: DeepSeek-R1 takes longer because it is a reasoning model that generates internal "Chain of Thought" tokens before producing its final answer. This requires more CPU cycles and sequential processing than the standard "next-token" prediction used by the smaller Llama and SmolLM models.

7. Response Quality Comparison

Model Name	Critique Accuracy(1-5)	Reasoning Depth(1-5)
Llama 3.2	3.3	2.3
DeepSeek-R1	3.3	3.0
SmolLM2	4.0	3.0